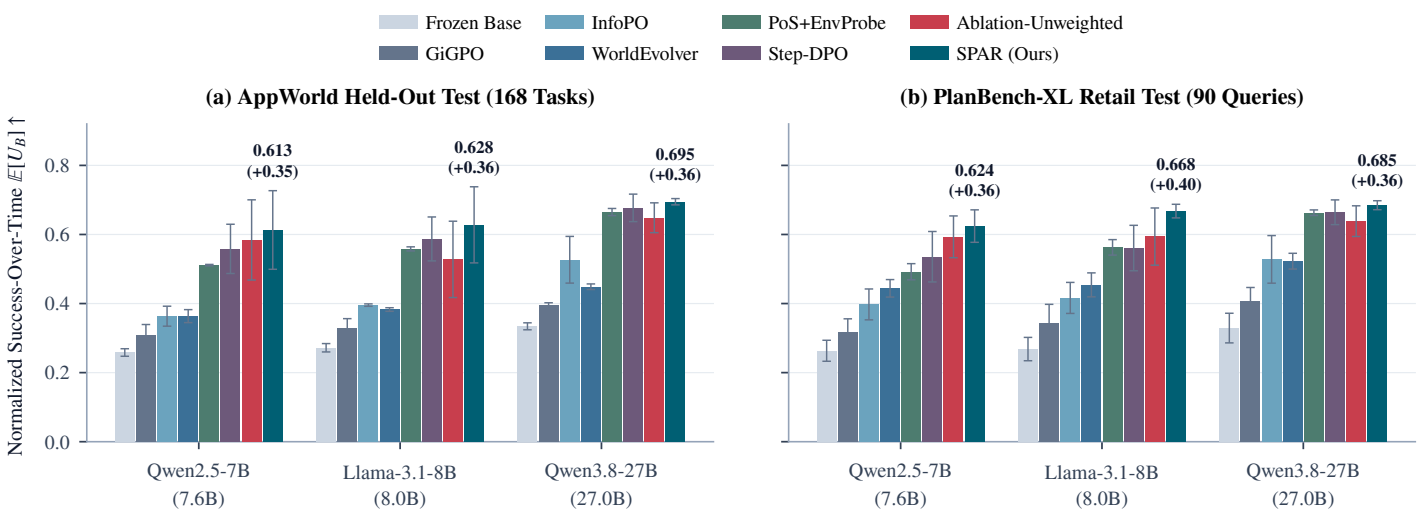
Frozen Base
InfoPO
PoS+EnvProbe
Ablation-Unweighted
GiGPO
WorldEvolver
Step-DPO
SPAR (Ours)
(a) AppWorld Held-Out Test (168 Tasks)
(b) PlanBench-XL Retail Test (90 Queries)
Normalized Success-Over-Time $\mathbb{E}[U_B] \uparrow$
0.0
0.2
0.4
0.6
0.8
0.613 (+0.35)
0.628 (+0.36)
0.695 (+0.36)
0.624 (+0.36)
0.668 (+0.40)
0.685 (+0.36)
Qwen2.5-7B (7.6B)
Llama-3.1-8B (8.0B)
Qwen3.8-27B (27.0B)